# Failure of exterior observability for a Schrödinger equation with slower surface dispersion

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**Abstract**

We consider a Schrödinger field on a planar annulus with a dynamic inner boundary and a Dirichlet outer boundary. When the surface dispersion coefficient is one half of the bulk coefficient, observation of the outer normal derivative fails to control the initial energy at every positive time. We construct smooth solutions with unit initial energy and exponentially small observation on each fixed time interval. High angular frequencies produce modes concentrated near the inner boundary. A variational construction on a short collar, followed by a comparison in the graph norm of the evolution operator, turns these localized modes into exact solutions on the full annulus.

## 1 The observation problem

Dynamic boundary conditions allow the trace of a dispersive field to evolve along the boundary while remaining coupled to the bulk through the normal derivative. An exterior observability estimate asks whether the initial energy can be bounded by the flux measured on a separate boundary component. Such estimates are central to boundary control of the coupled system [1].

Let

$$\Omega = \{x \in \mathbb{R}^2 : 1 < |x| < 2\}, \qquad \Gamma_1 = \{|x| = 1\}, \qquad \Gamma_0 = \{|x| = 2\}.$$

We study complex-valued solutions $Z = (z, z_\Gamma)$ of

$$\begin{cases} iz_t + \Delta z = 0 & \text{in } \Omega, \\ z = 0 & \text{on } \Gamma_0, \\ z_\Gamma = z|_{\Gamma_1}, & \\ iz_{\Gamma,t} + \frac{1}{2}\Delta_\Gamma z_\Gamma - \partial_\nu z = 0 & \text{on } \Gamma_1. \end{cases} \tag{1}$$

Here $\nu$ is outward from the annulus, so it points towards the origin on $\Gamma_1$, and $\Delta_\Gamma$ is the Laplace–Beltrami operator on the inner unit circle. The energy is

$$\|Z\|_{\mathcal{E}}^2 = \int_\Omega |\nabla z|^2 \, dx + \frac{1}{2}\int_{\Gamma_1} |\nabla_\Gamma z_\Gamma|^2 \, dS. \tag{2}$$

The question in [4] is whether, for every $T > 0$, a finite constant $C_T$ satisfies

$$\|Z(0)\|_{\mathcal{E}}^2 \le C_T \int_0^T \int_{\Gamma_0} |\partial_\nu z|^2 \, dS \, dt \tag{3}$$

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for every smooth compatible solution.

Mercado and Morales [1, author preprint, Theorem 1.9 and equation (3.20)] prove boundary controllability when the surface coefficient $\delta$ is larger than the bulk coefficient $d$. Their Carleman argument obtains a surface gradient term with coefficient $d(\delta - d)$, and does not transfer unchanged to the present regime $d = 1$, $\delta = 1/2$. The slower-surface Schrödinger regime is identified as an open question in [2, Section 5.1], separately from the corresponding negative result for waves.

**Theorem 1.** *There are constants $c > 0$ and $N \in \mathbb{N}$, and smooth compatible solutions $Z_n$ of (1), indexed by integers $n \geq N$, such that*

$$\|Z_n(0)\|_{\mathcal{E}}^2 = 1, \qquad \int_0^T \int_{\Gamma_0} |\partial_\nu z_n|^2 \, dS \, dt \leq K_T n^6 e^{-2cn} \quad \text{for every } T > 0,$$

*where $K_T < \infty$ is independent of $n$. Consequently (3) fails for every $T > 0$.*

For angular frequency $n$, the surface energy level is $n^2/2$, whereas the bulk angular potential near the inner circle is $n^2/r^2$. The difference confines a low surface mode to a thin inner layer. We construct this mode on a collar where the whole bulk potential lies above the surface level, then cut it off before extending it to the full annulus. A graph norm estimate controls the outer derivative of the error in the exact evolution. This permits the construction without isolating an eigenvalue of the full annulus from its bulk spectrum.

## 2 Angular reduction and the outer trace

Write $\phi_n(\theta) = (2\pi)^{-1/2} e^{in\theta}$. For $1 < R \leq 2$, use the Hilbert space

$$\mathcal{H}_R = L^2((1, R), r\, dr) \oplus \mathbb{C}, \qquad \|(v, \alpha)\|_{\mathcal{H}_R}^2 = \int_1^R r|v|^2 \, dr + |\alpha|^2,$$

with inner product linear in the second argument. On the form domain

$$\mathcal{V}_R = \{(v, v(1)) : v \in H^1(1, R), \ v(R) = 0\}$$

define

$$q_{n,R}(v) = \int_1^R \left( r|v'|^2 + \frac{n^2}{r}|v|^2 \right) dr + \frac{n^2}{2}|v(1)|^2. \tag{4}$$

All endpoint values use the absolutely continuous Sobolev representatives. The associated sesquilinear form is the polarization of (4). It is densely defined: approximate a bulk function by smooth functions supported in the interior, and approximate an arbitrary surface coordinate by a bump at 1 with shrinking bulk mass. For fixed $n, R$, its form norm is equivalent to the $H^1$ norm, and both trace constraints persist under convergence. Thus it is closed and nonnegative.

The representation theorem for closed forms [3, Theorem 2.14] gives a self-adjoint operator $A_{n,R}$. Its domain and action are

$$\begin{aligned} D(A_{n,R}) &= \{(v, v(1)) : v \in H^2(1, R), \ v(R) = 0\}, \\ A_{n,R}(v, v(1)) &= \left( L_n v, \frac{n^2}{2} v(1) - v'(1) \right), \\ L_n v &= -v'' - r^{-1} v' + n^2 r^{-2} v. \end{aligned} \tag{5}$$

Indeed, the weak bulk equation for a represented form functional gives $(rv')' \in L^2$, hence $v \in H^2$. Integration by parts then identifies the surface component:

$$\int_1^R r\overline{u'}v'\,dr = -\int_1^R \overline{u}\,(rv')'\,dr - \overline{u(1)}v'(1), \qquad u(R) = 0.$$

Conversely this identity shows that each state in the displayed $H^2$ domain represents its form functional. There is no additional Robin condition in this operator domain.

Put $A_n = A_{n,2}$. Its unitary group $e^{-itA_n}$ preserves $D(A_n)$ and commutes with $A_n$ [3, Theorem 5.1]. If $U(t) = (v(t), v(t,1))$ solves $iU_t = A_nU$, then $z(t,r,\theta) = v(t,r)\phi_n(\theta)$ solves (1). In particular the boundary equation is

$$iv_t(t,1) = \frac{n^2}{2}v(t,1) - v_r(t,1),$$

which has the sign prescribed by $\partial_\nu = -\partial_r$ at 1. The angular normalization gives the exact identities

$$\|Z\|_{\mathcal{E}}^2 = q_{n,2}(v), \qquad \int_{\Gamma_0} |\partial_\nu z|^2\,dS = 2|v_r(2)|^2. \tag{6}$$

The factor 2 comes from $dS = 2\,d\theta$ on the outer circle.

**Lemma 2.** *There is a constant $C$, independent of $n \geq 1$, such that*

$$|v'(2)| \leq C\left((1+n^2)\,\|V\|_{\mathcal{H}_2} + \|A_nV\|_{\mathcal{H}_2}\right), \qquad V = (v, v(1)) \in D(A_n).$$

*Proof.* Positivity of the form and Cauchy–Schwarz give

$$\|v'\|_{L^2(1,2)}^2 \leq q_{n,2}(v) = \langle V, A_nV\rangle \leq \|V\|_{\mathcal{H}_2}\,\|A_nV\|_{\mathcal{H}_2}\,.$$

The bulk part of (5) implies

$$\|v''\|_{L^2(1,2)} \leq C\left(n^2\,\|V\|_{\mathcal{H}_2} + \|v'\|_{L^2(1,2)} + \|A_nV\|_{\mathcal{H}_2}\right).$$

For $g \in H^1(1,2)$, integration yields

$$g(2) = \int_1^2 g(r)\,dr + \int_1^2 (r-1)g'(r)\,dr.$$

Apply this identity to $g = v'$, and use $\sqrt{ab} \leq (a+b)/2$, to obtain the claim. □

## 3 A mode confined to an inner collar

Fix $b = 5/4$, so $b^{-2} = 16/25 > 1/2$. The constants below may depend on $b$ and fixed cutoffs, but are independent of $n$. Their values can increase from one estimate to the next.

**Lemma 3.** *For each $n \geq 1$, there are $\lambda_n \in \mathbb{R}$ and a real $u_n \in C^\infty[1,b]$, positive on $[1,b)$, satisfying*

$$\begin{gathered} L_nu_n = \lambda_nu_n, \qquad u_n(b) = 0, \qquad u_n'(1) = \left(\frac{n^2}{2} - \lambda_n\right)u_n(1), \\ \int_1^b ru_n^2\,dr + u_n(1)^2 = 1, \end{gathered} \tag{7}$$

*and*

$$\frac{n^2}{2} \leq \lambda_n \leq \frac{n^2}{2} + Cn. \tag{8}$$

*Proof.* Minimize $q_{n,b}$ over the unit sphere in $\mathcal{V}_b$, with the $\mathcal{H}_b$ norm. A minimizing sequence has bounded derivative norm, bounded value at 1, and is uniformly bounded and equicontinuous. It has a uniformly convergent subsequence and a weakly convergent derivative subsequence. For the latter extraction and lower semicontinuity one may use [3, Lemmas 1.12–1.13]. The identity $v(r) - v(1) = \int_1^r v'$ identifies the limit derivative. Uniform convergence retains the trace at $b$ and the unit mass; weak lower semicontinuity applies to $\sqrt{r}\, v'$. The minimum is therefore attained.

Replacing a minimizer by its modulus does not increase the derivative energy. One obtains $|(|v|)'| \le |v'|$ by regularizing the modulus as $(|v|^2 + \varepsilon^2)^{1/2}$ and passing to a weak derivative limit. We may choose a nonnegative real minimizer $u_n$. Variation of $q_{n,b} - \lambda_n \|\cdot\|^2_{\mathcal{H}_b}$ in real and imaginary directions gives its weak eigenvector equation, hence (7). The scalar ODE gives smoothness. If $u_n$ vanished at an interior point, its derivative there would vanish too; uniqueness for the linear ODE would force $u_n = 0$. If $u_n(1) = 0$, the boundary equation gives the same contradiction. Thus it is positive on $[1, b)$.

Because $r^{-2} \ge b^{-2} > 1/2$, the form satisfies $q_{n,b}(v) \ge (n^2/2)\, \|(v, v(1))\|^2_{\mathcal{H}_b}$, giving the lower bound in (8). For the upper bound take a fixed smooth $\eta$, with $0 \le \eta \le 1$, equal to 1 near 1 and zero near $b$. The trial function $\eta(r)e^{-n(r-1)}$ has squared mass at least 1, and its form is at most

$$\frac{n^2}{2} + bn + \frac{b\,\|\eta'\|^2_\infty}{n} + \frac{n}{2}.$$

Here $\int_1^b e^{-2n(r-1)} dr \le (2n)^{-1}$ bounds both bulk terms. The variational characterization gives the upper bound. □

**Lemma 4.** *For all sufficiently large $n$,*

$$u_n(1)^2 \ge \frac{1}{2}.$$

*On $I = [9/8, 6/5]$ there is a fixed $c > 0$ such that*

$$|u_n| \le Ce^{-cn}, \qquad |u_n'| + |u_n''| \le Cn^2 e^{-cn}, \qquad |u_n'''| \le Cn^4 e^{-cn}. \tag{9}$$

*Proof.* Set $y_n(r) = r^{1/2} u_n(r)$. The radial equation becomes

$$y_n'' = Q_n y_n, \qquad Q_n(r) = \frac{n^2 - 1/4}{r^2} - \lambda_n. \tag{10}$$

By (8) and $b^{-2} - 1/2 = 7/50$, for all large $n$,

$$Q_n(r) \ge \kappa^2 n^2 \quad (1 \le r \le b), \qquad \kappa^2 = \frac{7}{100}.$$

Compare $y_n$ with

$$h_n(r) = y_n(1) \frac{\sinh(\kappa n(b - r))}{\sinh(\kappa n(b - 1))}.$$

The function $y_n - h_n$ has zero endpoint values and

$$(-\partial_r^2 + \kappa^2 n^2)(y_n - h_n) = (\kappa^2 n^2 - Q_n) y_n \le 0.$$

A positive interior maximum is impossible. Since $y_n(1) = u_n(1) \le 1$, we obtain

$$0 \le y_n(r) \le \frac{e^{-\kappa n(r-1)}}{1 - e^{-2\kappa n(b-1)}} \le Ce^{-\kappa n(r-1)}. \tag{11}$$

Consequently

$$\int_1^b r u_n^2 \, dr = \int_1^b y_n^2 \, dr \le \frac{C}{n},$$

and the normalization in (7) gives $u_n(1)^2 \ge 1 - C/n \ge 1/2$.

To estimate derivatives, take $h = 1/40$. For $r \in I$, the interval $[r, r+h]$ lies inside $(1, b)$ and remains at distance at least $1/8$ from 1. On these intervals $Q_n, Q_n' = O(n^2)$, so (10) and (11) bound $y_n''$. The identity

$$y_n'(r) = \frac{y_n(r+h) - y_n(r)}{h} - \frac{1}{h}\int_r^{r+h} (r + h - s) y_n''(s) \, ds$$

then bounds $y_n'$ by $Cn^2 e^{-\kappa n/8}$. Finally $y_n''' = Q_n' y_n + Q_n y_n'$ is bounded by $Cn^4 e^{-\kappa n/8}$. Differentiating $u_n = r^{-1/2} y_n$ gives (9) with $c = \kappa/8$. □

## 4 Exact solutions with small exterior observation

*Proof of Theorem 1.* Choose a fixed smooth cutoff $\chi$, with $0 \le \chi \le 1$, equal to 1 on $[1, 9/8]$ and zero on $[6/5, b]$. Define $w_n = \chi u_n$ on $[1, b]$, extend it by zero to $[1, 2]$, and set

$$W_n = (w_n, u_n(1)) \in \mathcal{H}_2.$$

The inner boundary equation is unchanged by this cutoff. From (7),

$$F_n = (A_n - \lambda_n) W_n = (f_n, 0), \qquad f_n = -2\chi' u_n' - (\chi'' + r^{-1}\chi') u_n. \tag{12}$$

The function $f_n$ is smooth and supported in $I$, away from both actual boundaries. Lemma 4 gives

$$\|F_n\|_{\mathcal{H}_2} \le Cn^2 e^{-cn}, \qquad \|A_n F_n\|_{\mathcal{H}_2} \le Cn^4 e^{-cn}. \tag{13}$$

For the second bound, two derivatives of $f_n$ involve profile derivatives only up to $u_n'''$. The potential term contributes $n^2 r^{-2} f_n$; the surface component of $A_n F_n$ is zero. The retained surface mass also gives

$$q_{n,2}(w_n) \ge \frac{n^2}{2} u_n(1)^2 \ge \frac{n^2}{4}. \tag{14}$$

Every successive operator image of $F_n$ remains smooth and compactly supported in the interior. Thus $F_n$ belongs to every $D(A_n^k)$. Since $W_n \in D(A_n)$, the identity $A_n W_n = \lambda_n W_n + F_n$ proves inductively that $W_n$ also belongs to every $D(A_n^k)$. Define the exact evolution

$$U_n(t) = e^{-itA_n} W_n = (v_n(t), v_n(t, 1)).$$

The unitary group preserves all power domains and gives arbitrary time regularity in their graph norms. The bulk differential equation in (5) successively gives $H^{2k}$ spatial regularity from membership in $D(A_n^k)$. Finite-interval Sobolev representatives then give a smooth radial function. The power-domain trace constraints supply all boundary compatibility identities. Multiplying by the normalized angular factor $\phi_n(\theta) = (2\pi)^{-1/2} e^{in\theta}$ gives the smooth compatible solution $\widetilde{z}_n(t, r, \theta) = v_n(t, r)\phi_n(\theta)$ of (1), together with its inner trace.

Compare it with the approximate stationary evolution and write

$$D_n(t) = U_n(t) - e^{-it\lambda_n} W_n.$$

Differentiating $e^{-i(t-s)A_n}e^{-is\lambda_n}W_n$ in the graph norm gives Duhamel's formula

$$D_n(t) = -i\int_0^t e^{-i(t-s)A_n}e^{-is\lambda_n}F_n\,ds.$$

Unitarity and commutation with $A_n$ yield

$$\|D_n(t)\|_{\mathcal{H}_2} \le t\,\|F_n\|_{\mathcal{H}_2}\,, \qquad \|A_nD_n(t)\|_{\mathcal{H}_2} \le t\,\|A_nF_n\|_{\mathcal{H}_2}\,.$$

As $w_n$ vanishes near 2, its outer derivative is zero. Lemma 2 and (13) therefore imply

$$|\partial_r v_n(t,2)| \le Ctn^4e^{-cn} \qquad (t \ge 0).$$

Using the exact normalization (6), we find

$$\int_0^T\int_{\Gamma_0} |\partial_\nu \widetilde{z}_n|^2\,dS\,dt = 2\int_0^T |\partial_r v_n(t,2)|^2 dt \le K_T n^8 e^{-2cn}.$$

Finally divide $\widetilde{z}_n$ and its trace by $\sqrt{q_{n,2}(w_n)}$, which is nonzero by (14). The resulting solution $Z_n$ has initial energy one and observation at most $K_T n^6 e^{-2cn}$. This tends to zero for every fixed $T > 0$, contradicting (3) for any solution-independent $C_T$. □

**AI use.** This manuscript was developed with OpenAI Codex (gpt-6.1-sol), including the proof construction, literature checks and writing. The checks performed during preparation were AI checks; independent mathematical review is requested.

## References


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